

(向量值函数在空间中运动)

$$1. \text{Proj}_v u = \frac{u \cdot v}{|v|^2} \cdot \frac{v}{|v|}$$

$$u \times v = |u| |v| \sin \theta n \quad (\theta \text{ 为夹角})$$

$$\textcircled{1} (ru) \times (sv) = (rs) (u \times v) \quad \textcircled{2} u \times v = -(v \times u)$$

$$\textcircled{3} u \times (v \times w) = (u \cdot w)v - (u \cdot v)w$$

$$\text{Triple scalar product } (u \times v) \cdot w = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

$$\text{点到直线的距离 } d = \frac{|\vec{PS} \times \vec{v}|}{|v|}$$

$$\text{点到平面的距离 } d = \frac{|\vec{PS} \cdot \vec{n}|}{|\vec{n}|}$$

$$\text{Ellipsoid } \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$\text{Elliptical paraboloid } \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$$

$$\text{Elliptical cone } \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$$

$$\text{Hyperboloid of 1-Sheet } \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

$$\text{Hyperboloid of 2-Sheet } \frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\text{Hyperbolic paraboloid } \frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}, c > 0$$

$$\text{Arc Length Formula: } L = \int_a^b |v| dt$$

$$T = \frac{v}{|v|} = \frac{dv}{ds}$$

$$\text{Principal unit normal vector: } N = \frac{\frac{dT}{ds}}{|\frac{dT}{ds}|}$$

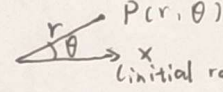
$$\text{Binormal vector: } B = T \times N$$

$$\text{Curvature: } \kappa = \left| \frac{dT}{ds} \right| = \frac{|v \times a|}{|v|^3}$$

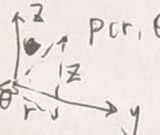
$$\text{Torsion: } \tau = -\frac{dB}{ds} \cdot N = \frac{\begin{vmatrix} x & y & z \\ \dot{x} & \dot{y} & \dot{z} \\ \ddot{x} & \ddot{y} & \ddot{z} \end{vmatrix}}{|v \times a|^2}$$

2. 坐标系

① Polar Coordinates. 极坐标

↺ 逆时针. $P(r, \theta) \rightarrow$ 同一点极坐标的坐标表示不唯一.


② Cylindrical Coordinates. 柱面坐标

$r \geq 0$. 

$$\begin{cases} \vec{r} = r \cdot \vec{u}_r + z \cdot \vec{k} \\ \vec{v} = \dot{r} \cdot \vec{u}_r + r \dot{\theta} \cdot \vec{u}_\theta + \dot{z} \cdot \vec{k} \\ \vec{a} = (\ddot{r} - r\dot{\theta}^2) \cdot \vec{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \cdot \vec{u}_\theta + \ddot{z} \cdot \vec{k} \end{cases}$$

3. 偏导

多元函数极限存在的条件: 从各个方向上极限都相等 [Two Path Test].

连续: 极限存在且等于函数值.

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta, |f(x,y) - L| < \epsilon, \text{ then } \lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L.$$

[Continuity of Composition]. f, g 在 (x_0, y_0) 连续 $\Rightarrow f \circ g$ 在 (x_0, y_0) 连续.

$$\text{1阶偏导: } \frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h, y_0) - f(x_0, y_0)}{h} = \frac{d}{dx} f(x, y_0)$$

$$\text{2阶偏导: } \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0, y_0+h) - f(x_0, y_0)}{h} = \frac{d}{dy} \frac{\partial f}{\partial x}(x_0, y)$$

(x_0, y_0) 的切平面.

2阶偏导: [Mixed Derivative Theorem], f_x, f_y, f_{xy}, f_{yx} 都连续 $\Rightarrow f_{xy}(a,b) = f_{yx}(a,b)$.

可微性: f 在 (x_0, y_0) 可微 $\Leftrightarrow$ ① $f_x(x_0, y_0), f_y(x_0, y_0)$ 存在.

$\Leftrightarrow$ ② $z = f(x, y)$

其中 $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$ 满足

$$\Delta z = f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y$$

当 $\Delta x, \Delta y \rightarrow 0$ 时 $\epsilon_1, \epsilon_2 \rightarrow 0$.

f 在 D_f 上可微 $\Leftrightarrow f$ 光滑. Power Series. $A(x) = \sum_{n=0}^{\infty} a_n x^n, B(x) = \sum_{n=0}^{\infty} b_n x^n$ 在 $|x| < R$ 绝对收敛, 则

$\sum_{n=0}^{\infty} C_n x^n = (\sum_{n=0}^{\infty} a_n x^n) (\sum_{n=0}^{\infty} b_n x^n)$ 在 $|x| < R$ 绝对收敛, 其中

$C_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \dots = \sum_{k=0}^n a_k b_{n-k}$

2. $\sum_{n=0}^{\infty} a_n x^n$ 在 $|x| < R$ 绝对收敛, f 连续 $\Rightarrow \sum_{n=0}^{\infty} a_n (f(x))^n$ 在 $|f(x)| < R$ 绝对收敛.

Limit: $f: \mathbb{R}^n \rightarrow \mathbb{R}, \forall \epsilon > 0, \exists \delta > 0$ s.t. $\forall t \in D_f$

$$|f(x) - L| < \epsilon \text{ whenever } 0 < |x - x_0| < \delta \text{ then } \lim_{x \rightarrow x_0} f(x) = L.$$

Continuity: f 在 x_0 连续 $\Leftrightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0)$

Fourier Series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L})$

① 周期性, Fundamental period: $T = \frac{2L}{n}$

Common period: $T = 2L$.

② 正交性: $(u, v) = \int_a^b u(x) v(x) dx$

u, v 在 $x \in [a, b]$ 上正交

$\int_a^b u(x) v(x) dx = 0$.

Tangential and normal scalar components of acceleration

$$\vec{a} = a_t \vec{T} + a_n \vec{N} \quad \textcircled{3} \text{ 求 Fourier coefficients } \{a_n, b_n\}$$

$$a_t = \frac{dv}{dt} \quad \text{Euler-Fourier Formulas}$$

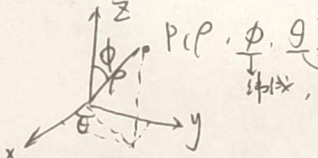
$$a_n = \kappa |v|^2 \quad a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \left(\frac{n\pi}{L} x \right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \left(\frac{n\pi}{L} x \right) dx$$

$$\text{Polar} \Leftrightarrow \text{Cartesian: } \begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ r^2 = x^2 + y^2 \\ \tan \theta = \frac{y}{x} \end{cases}$$

③ Spherical Coordinates. 球面坐标



Spherical $\Leftrightarrow$ Cartesian $\Leftrightarrow$ Cylindrical

$$\begin{cases} r = \rho \sin \phi, & x = r \cos \theta = \rho \sin \phi \cos \theta \\ z = \rho \cos \phi, & y = r \sin \theta = \rho \sin \phi \sin \theta \\ \rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2} \end{cases}$$

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}$$

$$\textcircled{1} \text{ Chain Rule: } w = f(x, y, z), x = g(t), y = h(t)$$

$$z = m(t) \Rightarrow \frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

② Implicit Differentiation:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}, F(x, y) = 0$$

方向导: f 在 (x_0, y_0) 处单位方向 $\vec{u} = u_1 \vec{i} + u_2 \vec{j}$

的导数为

$$f'_u(x_0, y_0) = \lim_{s \rightarrow 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}$$

s 为 P 到 $\vec{u}$ 的弧长.

$$\left(\frac{df}{ds} \right)_{u, P}$$

梯度: $\nabla f(x_0, y_0) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (f_x, f_y)$

$$\Rightarrow D_u f = \left(\frac{\partial f}{\partial s} \right)_{u, P} = \nabla f(x_0, y_0) \cdot \vec{u}$$

$$D_u f = \left(\frac{\partial f}{\partial s} \right)_{u, P} = \nabla f \cdot \vec{u} = |\nabla f| \cdot \cos \theta$$

f 在方向 u 上的变化量: $df = \nabla f(x_0, y_0) \cdot \vec{u}$

3. $f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n$ 在 $|x-a| < R$ 绝对收敛

$\Rightarrow \int_a^b f(x) dx$ 也在 $|x-a| < R$ 绝对收敛

证明: ∇f 与等值线正交.

[证]. 设 $f = (x, y)$. f 的等值线为 $r(t) = g(t)\vec{i} + h(t)\vec{j}$

则 f 的 C 等值线为 $f(g(t), h(t)) = C$.

$$\frac{d}{dt} f(g(t), h(t)) = \frac{d}{dt} C \Rightarrow \frac{dx}{dt} \frac{df}{dx} + \frac{dy}{dt} \frac{df}{dy} = 0$$

$$\Rightarrow (f_x, f_y) \cdot \left(\frac{dx}{dt}, \frac{dy}{dt} \right) = 0 \Rightarrow \nabla f \cdot \vec{r}' = 0$$

$\Rightarrow \nabla f$ 与等值线的切线正交 $\Rightarrow \nabla f$ 与等值线正交. 证.

[The Derivative along a Path]. $\frac{d}{dt} f(r(t)) = \nabla f(r(t)) \cdot r'(t)$.

[线性化估计]. f 的 1, 2 阶偏导连续, 则 f 在 (x_0, y_0) 的 Linearization: $L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

误差为 $E(x, y) = f(x, y) - L(x, y)$, 满足 $|E(x, y)| \leq \frac{1}{2} M \cdot (|x - x_0| + |y - y_0|)^2$, 其中 $M = \max \{|f_{xx}|, |f_{xy}|, |f_{yy}|\}$

[First Derivative Test]. 对每点且 $\nabla f = 0 \Rightarrow$ 关键点 critical point, critical point $\Rightarrow \nabla f = 0$ 或 f_x, f_y (其中一) DNE.

[Second Derivative Test]. f 的 1, 2 阶偏导连续.

① $f_{xx} < 0, f_{xx}f_{yy} - f_{xy}^2 > 0 \Rightarrow$ maximum.

② $f_{xx} > 0, f_{xx}f_{yy} - f_{xy}^2 > 0 \Rightarrow$ minimum.

③ $f_{xx}f_{yy} - f_{xy}^2 < 0 \Rightarrow$ saddle point.

④ $f_{xx}f_{yy} - f_{xy}^2 = 0 \Rightarrow$ unknown.

$$\text{其中 } f_{xx}f_{yy} - f_{xy}^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = |H_f|$$

有界闭上的最大/小值: 在边界点或关键点上.

Taylor Formula.

设 f 有 $n+1$ 阶连续偏导, 则 f 在 (x_0, y_0) 的 Taylor 展开:

$$f(a+h, b+k) = f(a, b) + (hf_x + kf_y) + \frac{1}{2!} (h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + \frac{1}{3!} (h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}) + \dots + \frac{1}{n!} (h^n f_{x^n} + \dots + k^n f_{y^n})$$

$$f(a, b) + \frac{(n+1)!}{(n+1)!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^n f(a+h, b+k)$$

前 n 项估计多项式, 最后一项估计误差.

$n=1$: 线性估计 Linearization. 最后一项 $= E(x, y)$

$n=2$: quadratic approximation.

4. (Multiple integral). $\iiint F(x, y, z) dx dy dz$

Average value of F over $D = \frac{1}{\text{Volume of } D} \iiint F dV$

rectangular. $f(x, y, z)$. 找体积: ① 对 z 积分 (作 3 维图)

cyindrical. $f(r, \theta, z)$. $r dr d\theta dz$. ② 被积函数是 1.

Spherical. $f(\rho, \phi, \theta)$. $\rho^2 \sin \phi d\rho d\phi d\theta$

Substitution. $[x, y] \rightarrow [g(u, v), h(u, v)]$.

Jacobian determinant.

$$J(u, v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$

$$\iint_R f(x, y) dx dy = \iint_a^b f(g(u, v), h(u, v)) |J(u, v)| du dv$$

$$\iiint_D F(x, y, z) dx dy dz = \iiint_a^b F(g(u, v, w), h(u, v, w), k(u, v, w)) |J(u, v, w)| du dv dw$$

$f(x, y, z)$ 的参数方程: $r(t) = g(t)\vec{i} + h(t)\vec{j} + k(t)\vec{k}$.

f 在 $C: r(t) = g(t)\vec{i} + h(t)\vec{j} + k(t)\vec{k}$ 上的 Line Integral:

$$\int_C f(x, y, z) ds = \int_a^b f(g(t), h(t), k(t)) |\vec{r}'(t)| dt$$

Series and Error Estimation.

1. Integral. $a_k = f(k)$. 收敛 $\Rightarrow \downarrow, (V_N) \Rightarrow \sum a_n \rightarrow S$.

对 Remainder $R_n = S - S_n$ 满足

$$\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx$$

Partial sum S_n 满足

$$S_n + \int_{n+1}^{\infty} f(x) dx \leq S \leq \int_n^{\infty} f(x) dx + S_n$$

2. Alternating. $\sum_{n=1}^{\infty} (-1)^n u_n$ 满足 Alternating Series Test 的条件, 则

$$S_n = u_1 - u_2 + \dots + (-1)^n u_n$$

$$\Rightarrow |R_n| = |S - S_n| < |u_{n+1}|$$

对于 Absolute convergent series, 改变序列顺序收敛值不变

(Conditionally Convergent 无该性质)

对于 Vector Field F (线积为 line integral $= \int_C \vec{F} \cdot d\vec{s}$)

面积分 Surface integral $= \text{flux} = \iint_S \vec{F} \cdot \vec{n} dA = \iint_S \vec{F} \cdot \vec{n} |\vec{r}_u \times \vec{r}_v| du dv$

$\Rightarrow$ 等值线在 (x_0, y_0) 处的切线:

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 0, \quad (\nabla f \cdot \vec{v} = 0)$$

$\Rightarrow$ 考虑 f 在 (x_0, y_0) 处的切面, 有在 P_0 处的所有切线都与 ∇f 正交.

则等值面 $f(x, y, z) = C$ 在 (x_0, y_0, z_0) 处的切面:

$$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) = 0$$

该切面的法线: $\begin{cases} x = x_0 + f_x t \\ y = y_0 + f_y t \\ z = z_0 + f_z t \end{cases}$ (与 ∇f 平行)

[Method of Lagrange Multiplier] 拉格朗日乘数法.

$f(x, y, z)$ 在限制域 $g(x, y, z) = 0$ 上的极值点满足 $\nabla f = \lambda \nabla g$.

[Orthogonal Gradient Theorem]. $f(x, y, z)$ 在包含曲线 $C: r(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$ 的区域内可微.

设 (x_0, y_0, z_0) 是 f 的极大值/小值点, 则 ∇f 在 (x_0, y_0, z_0) 与 C 正交.

[证]. $\nabla f \cdot r'(t) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \cdot \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$

$$= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = \frac{df}{dt} = 0$$

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[证拉格朗日乘数法].

设 $f(x, y, z), g(x, y, z) = 0$ 可微, P_0 在等值面 $g(x, y, z) = 0$ 上

设 $g \neq 0$ 且 P_0 是 f 的极值点, 则对于等值面上任意穿过 P_0 的可微曲线, 在 P_0 有极大/小值.

$\Rightarrow \nabla f$ 与 g 在 P_0 的切向量正交, 而 ∇g 也与 g 在 P_0 处切向量正交 $\Rightarrow \nabla f \parallel \nabla g \Rightarrow \nabla f = \lambda \nabla g, \lambda$ 为拉格朗日乘子.

[证]. $\nabla f \cdot r'(t) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \cdot \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$

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曲面的 (正) 定向性 orientation: $\vec{r}_u \times \vec{r}_v$. 不是所有曲面都有定向性.

面元的 Surface integral $= \text{flux} = \iint_S \vec{F} \cdot \vec{n} dA = \iint_S \vec{F} \cdot \vec{n} |\vec{r}_u \times \vec{r}_v| du dv$